

NANCY GRACE ROMAN OBSERVATORY POINTING ANGLES

1. NOMENCLATURE AND CONVENTIONS

All vectors are represented by lowercase bolded letters ($\mathbf{v}$). Unit vectors are represented by lowercase bolded letters with hats ($\hat{\mathbf{v}}$ is the unit vector of $\mathbf{v}$). Vector norms are denoted by $\|\mathbf{v}\|$ such that $\mathbf{v} = \|\mathbf{v}\|\hat{\mathbf{v}}$. Position vectors are denoted as $\mathbf{r}_{B/A}$ indicating that this is the vector pointing from point A to point B .

Reference frames are represented by calligraphic capital letters ($\mathcal{I}$) and defined by a coordinate origin and a set of three mutually orthogonal unit vectors such that $\mathcal{I} = (O, \hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}})$ implies that O is fixed in frame $\mathcal{I}$ and $\hat{\mathbf{i}} \times \hat{\mathbf{j}} = \hat{\mathbf{k}}$. All reference frames are dextral.

Direction cosine matrices (DCMs) are defined as ${}^{\mathcal{B}}C^{\mathcal{I}}$ such that:

$$[\mathbf{v}]_{\mathcal{B}} = {}^{\mathcal{B}}C^{\mathcal{I}}[\mathbf{v}]_{\mathcal{I}},$$

where $[\mathbf{v}]_{\mathcal{B}}, [\mathbf{v}]_{\mathcal{I}}$ are the component representations of vector $\mathbf{v}$ in components of the $\mathcal{B}$ and $\mathcal{I}$ frame, respectively (these are strictly 3x1 column matrices while the DCM is a 3x3 orthonormal matrix). We define the three DCMs associated with counter-clockwise (CCW) rotations about reference frame axes as:

$$C_1(\theta) \triangleq \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{bmatrix} \quad C_2(\theta) \triangleq \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix} \quad C_3(\theta) \triangleq \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The projection of a vector $\mathbf{a}$ onto a plane orthogonal to direction $\hat{\mathbf{b}}$ is given by:

$$\mathbf{a}_{\text{proj}} = \mathbf{a} - (\mathbf{a} \cdot \hat{\mathbf{b}})\hat{\mathbf{b}},$$

where $\cdot$ represents the scalar dot product.

The angle between unit vectors $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ assuming counter-clockwise rotation about axis $\hat{\mathbf{z}}$ is:

$$\theta = \text{atan2}(\|\hat{\mathbf{x}} \times \hat{\mathbf{y}}\| \text{sign}(\det[\hat{\mathbf{x}} \ \hat{\mathbf{y}} \ \hat{\mathbf{z}}]), \hat{\mathbf{x}} \cdot \hat{\mathbf{y}})$$

where atan2 is the two-term arctangent operator, $\det$ is a matrix determinant, and sign is the mathematical sign operator.

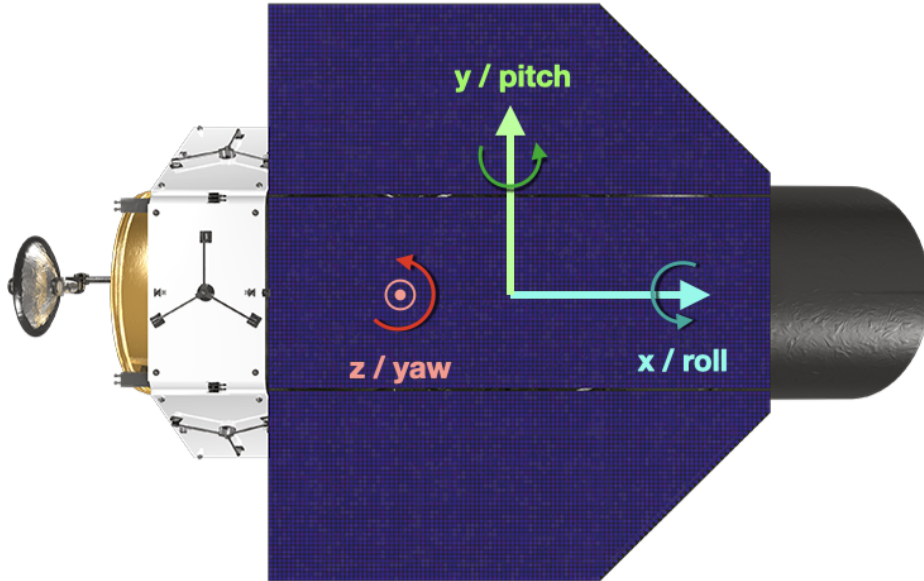


FIGURE 1. Roman body-fixed frame definition.

2. REFERENCE FRAME DEFINITION

We define the Roman body-fixed (or body-centered) reference frame ($\mathcal{B}$) via the mutually orthogonal unit vectors $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$, with the coordinate origin at the spacecraft's center of mass (COM; R) such that $\mathcal{B} = (R, \hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}})$. As in Figure 1, $\hat{\mathbf{z}}$ is orthogonal to the surface of the solar panels, $\hat{\mathbf{x}}$ is parallel to the center of the barrel, and $\hat{\mathbf{y}} = \hat{\mathbf{z}} \times \hat{\mathbf{x}}$.

We define an ecliptic inertial reference frame $\mathcal{I} = (O, \hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2, \hat{\mathbf{e}}_3)$ where O is the solar system barycenter, $\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2$ are in the ecliptic plane, and $\hat{\mathbf{e}}_3$ is the direction of ecliptic north. $\hat{\mathbf{e}}_1$ is first unit vector of ICRS (i.e., the vernal equinox direction or the first point of Aries). The specification of coordinates may be either via the mean equinox of epoch or true equinox of date, so long as all coordinates are self-consistent.

3. POINTING ANGLE DEFINITION

Roman pointing commands are described as a set of 3 ordered rotations about body-fixed axes in the order: $\hat{\mathbf{z}}, \hat{\mathbf{y}}, \hat{\mathbf{x}}$. The angles are labeled yaw, pitch, roll, respectively and denoted by ψ, θ, ϕ . The relationship between the orientation of Roman post re-pointing to its orientation pre re-pointing is therefore:

$$C_1(\phi)C_2(\theta)C_3(\psi).$$

4. ZERO-POINT DEFINITION

The zero-point for Roman pointing calculations is defined with $\hat{\mathbf{z}}$ pointing directly at the sun (denoted by $\odot$), such that $\hat{\mathbf{z}} \parallel \mathbf{r}_{\odot/R}$, where $\mathbf{r}_{\odot/R}$ is the vector from Roman's COM to the sun. In order to establish the relationship between the zero-point spacecraft orientation and our inertial frame, we initially orient Roman such that $\mathcal{B}$ is fully aligned with $\mathcal{I}$, as in Figure 2.

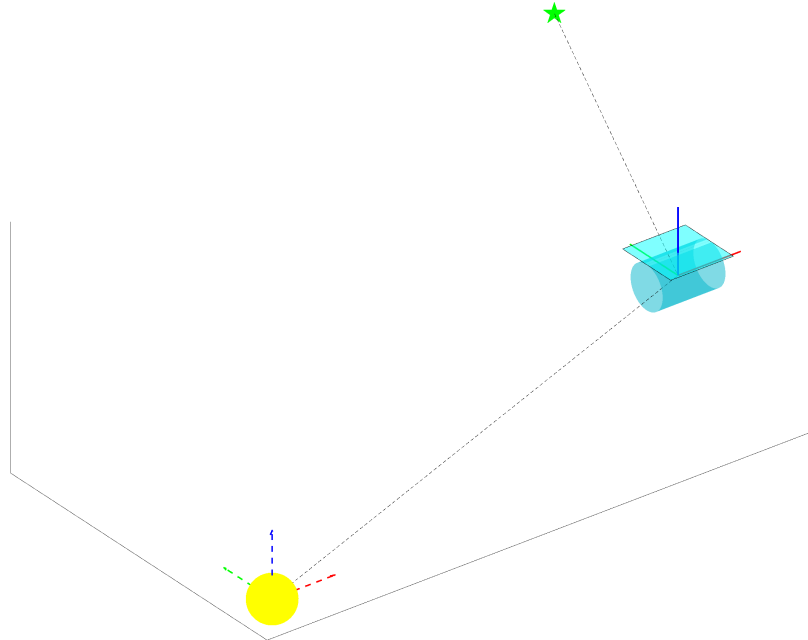


FIGURE 2. Initial reference orientation. In this, and all subsequent figures, the dashed, colored lines indicate frame $\mathcal{I}$ and solid, colored line indicate frame $\mathcal{B}$. The ordering of unit vectors in each case is denoted by red, green, blue (such that the solid blue line is the $\hat{\mathbf{z}}$ direction).

We can then compute two sequential rotations to align $\hat{\mathbf{z}}$ with $\mathbf{r}_{\odot/R}$. The first will be about $\hat{\mathbf{y}}$ by the angle between $\hat{\mathbf{e}}_3$ and the projection of $\mathbf{r}_{\odot/R}$ onto the $\hat{\mathbf{e}}_1 - \hat{\mathbf{e}}_3$ plane, which we'll denote as α_1 . The second rotation will be about $\hat{\mathbf{x}}$ by the angle between $\mathbf{r}_{\odot/R}$ and $\mathbf{z}$, which we'll call α_2 . Therefore, the DCM relating $\mathcal{I}$ to the zero-point orientation for a given observatory location is given by:

$${}^{\mathcal{B}}C_0^{\mathcal{I}} = C_1(\alpha_2)C_2(\alpha_1).$$

The zero-point orientation is shown in Figure 3.

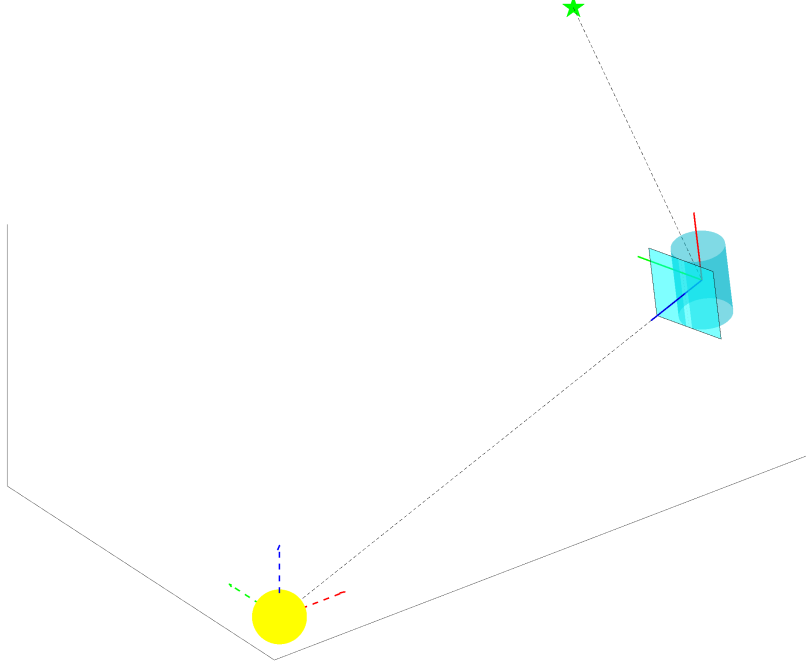


FIGURE 3. Zero-point orientation. $\hat{\mathbf{z}}$ is aligned with the sun-spacecraft line.

It is important to note that the zero-point orientation of $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ is entirely arbitrary and unimportant for the calculation of pitch and roll angles. Any change to the initial orientation of these directions would be accomplished via rotation about $\hat{\mathbf{z}}$ (yaw). As yaw is also the first element of the pointing angle set (such that yaw changes are executed first in aligning with a new target), this would represent two sequential yaw changes, which is equivalent to a single yaw maneuver. That is, $C_3(\theta_1)C_3(\theta_2) \equiv C_3(\theta_1 + \theta_2)$. The values of the pitch and roll angles are entirely decoupled from this, so changing the $\hat{\mathbf{x}}, \hat{\mathbf{y}}$ zero-point orientation only impacts the computed yaw maneuver, which is essentially unrestricted for Roman.

5. TARGET POINTING

Given a target star's barycentric ecliptic longitude and latitude at the epoch of observation (λ, β) , along with the stellar distance d , we can write:

$$[\mathbf{r}_{\star/O}]_{\mathcal{I}} = d \begin{bmatrix} \cos \lambda \cos \beta \\ \sin \lambda \cos \beta \\ \sin \beta \end{bmatrix}_{\mathcal{I}}.$$

We subtract the observatory's barycentric position from this vector to get the observatory-star look vector.

$$\mathbf{r}_{\star/G} = \mathbf{r}_{\star/O} - \mathbf{r}_{G/O}.$$

To align the boresight ($\hat{\mathbf{x}}$) with the look vector, we first rotate about the $\mathbf{z}$ axis by a yaw angle (ψ) equal to the angle between $\hat{\mathbf{x}}$ and the projection of $\mathbf{r}_{*/G}$ onto the $\hat{\mathbf{x}} - \hat{\mathbf{y}}$ plane. We then rotate about $\hat{\mathbf{y}}$ with a pitch angle (θ) equal to the angle between $\hat{\mathbf{x}}$ and $\hat{\mathbf{r}}_{*/G}$. The result of these two maneuvers is illustrated in Figure 4.

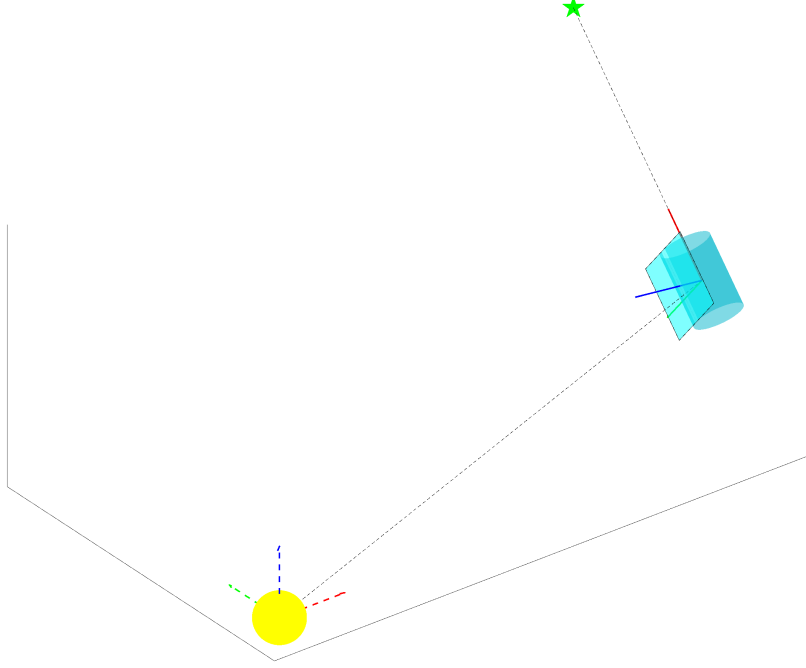


FIGURE 4. Zero-point orientation. $\hat{\mathbf{z}}$ is aligned with the sun-spacecraft line.

6. ROLL-PITCH COUPLING

Only the first two orientation degrees of freedom are required to align the observatory's boresight with the target look vector, leaving the remaining degree of freedom (roll) so far unused. The observatory carries a $\pm 13^\circ$ restriction on roll in order to ensure adequate panel illumination. The three Euler angles are computed independently of one another, so setting a value for roll won't change the previously computed value of the pitch angle. However, a non-zero roll angle *will* change the angle between the observatory-sun line and $\hat{\mathbf{z}}$, which may have an impact on thermal effects on the observatory due to solar heating.

For example: in the scenario shown in Figures 2-4, the computed pitch angle was approximately 20° . At the end of the yaw and pitch maneuvers, the angle between $\mathbf{r}_{\odot/G}$ and $\hat{\mathbf{z}}$ is exactly equal to the computed pitch value. However, if we roll the observatory about the boresight by 13° , the angle between $\mathbf{r}_{\odot/G}$ and $\hat{\mathbf{z}}$ will be approximately 24° —a change of nearly 4° .